

Chennai Civic Issues Reporting System

ADMIN DASHBOARD – CHENNAI CIVIC ISSUES REPORTING AND MANAGING SYSTEM

"CIVIC CONNECT"

Under the guidance
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INTRODUCTION

- Urban cities like Chennai face issues such as potholes, garbage overflow, water leakage and streetlight failures.
- Citizens often struggle to report these problems efficiently.
- Traditional complaint systems are slow and lack transparency.
- This project proposes a web-based civic issue reporting and management system.
- It allows citizens to easily report problems and administrators to monitor and resolve them.

PROBLEM STATEMENT

- Citizens find it difficult to report civic issues quickly.
- Manual complaint systems are slow and inefficient.
- Authorities may not receive accurate location or image evidence.
- Tracking complaint status is often not possible.
- There is a need for a centralized digital platform to report and manage civic problems.

MOTIVATION OF THE PROJECT

- Growth of smart city initiatives.
- Increasing use of smartphones and internet services.
- Need to improve communication between citizens and municipal authorities.
- Encourage public participation in city maintenance.
- Improve transparency and faster issue resolution.

OBJECTIVES OF THE PROJECT

- Develop an online platform for reporting civic issues.
- Allow users to upload images and descriptions of problems.
- Enable administrators to manage and resolve complaints.
- Provide tracking of complaint status.
- Improve communication between citizens and authorities.

LITERATURE REVIEW

Research Project / System	Year	Contributions	Limitations
CPGRAMS (Centralized Public Grievance Redress and Monitoring System)	2007	Provides a centralized national platform for citizens to submit grievances to government departments and track complaint status.	Requires registration with Aadhaar/OTP, lacks precise GPS-based location tagging, and has limited public transparency.
BBMP Sahaaya	2016	Enables citizens to report civic issues using a mobile application with photo upload and category-based complaint submission.	Requires installation of a mobile app and offers limited public dashboards for monitoring complaints.
FixMyStreet	2007	Introduces a map-based civic reporting platform with public visibility of complaints, encouraging community participation.	High chances of duplicate complaints and limited prioritization or grouping of similar issues.
SeeClickFix	2008	Provides a civic engagement platform with public issue feeds, mobile access, and integration with municipal work-order systems.	Commercial paid platform that may not be affordable for smaller municipalities and developing regions.
Chennai Civic Issues Reporting System (Proposed)	2026	Offers a web-based civic issue reporting system with GPS location tagging, role-based dashboards, real-time tracking, and analytics visualization.	Currently limited to web platform and requires internet connectivity for real-time updates.

LIMITATIONS OF EXISTING SYSTEMS

- Manual complaint handling in many municipalities.
- Limited real-time updates for citizens.
- Lack of centralized and digital complaint tracking.
- Poor communication between citizens and authorities.
- Limited data analysis of civic issues.

PROPOSED SYSTEM

- A web-based civic issue reporting platform.
- Citizens can report issues with description and images.
- Complaints are stored in a centralized database.
- Admin dashboard monitors and manages reported issues.
- Users can track the status of their complaints.

SYSTEM ARCHITECTURE

- User Interface (Citizen Portal)
- Backend Server (Node.js / Express)
- Database (MongoDB)
- Admin Dashboard
- Issue Management and Notification Modules

MODULES OF THE SYSTEM

- User Module – Allows users to submit complaints.
- Image Processing Module – Stores uploaded issue images.
- Database Module – Stores issue data and user details.
- Admin Monitoring Module – Displays reported issues.

MODULES OF THE SYSTEM

- Notification system for status updates.
- Admin alerts for newly reported issues.
- Tracking system for complaint progress until resolution.

SYSTEM WORKFLOW

- User logs into the system.
- User reports an issue with description and image.
- Complaint is stored in the database.
- Admin views complaints through the dashboard.
- Admin updates or resolves the issue.
- User tracks complaint status.

ALGORITHMS USED

- Authentication algorithm for user login.
- CRUD operations for complaint management.
- Database query processing for issue tracking.
- Image upload and storage handling.

Technology Stack

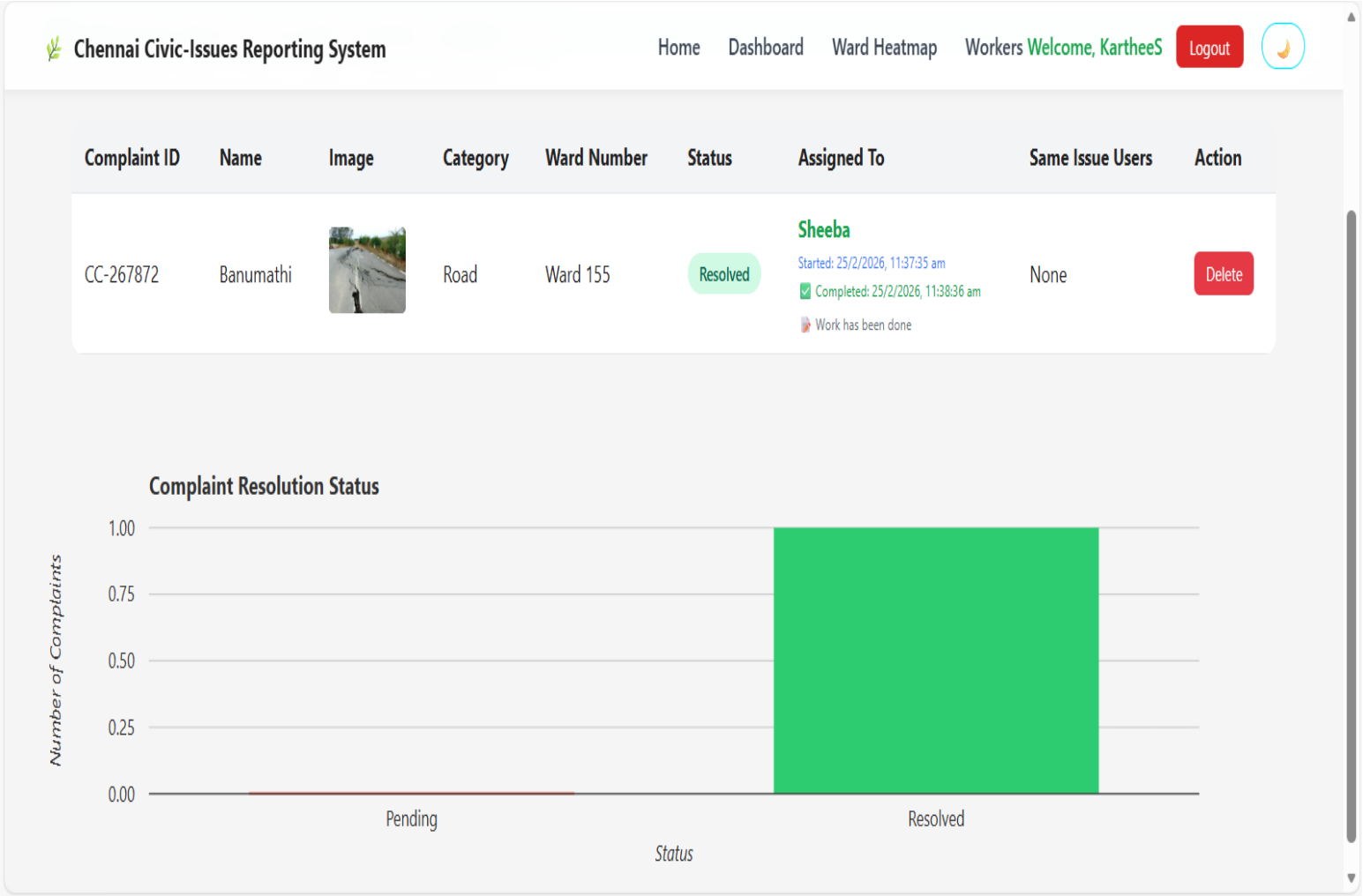
- Frontend: HTML, CSS, JavaScript
- Backend: Node.js and Express.js
- Database: MongoDB Compass Local Storage
- Development Tools: Visual Studio Code and GitHub

SYSTEM REQUIREMENTS

- Hardware: Computer or smartphone with internet connection.
- Software: Web browser, Node.js runtime.
- Database: MongoDB Compass
- Development Tools: VS Code.

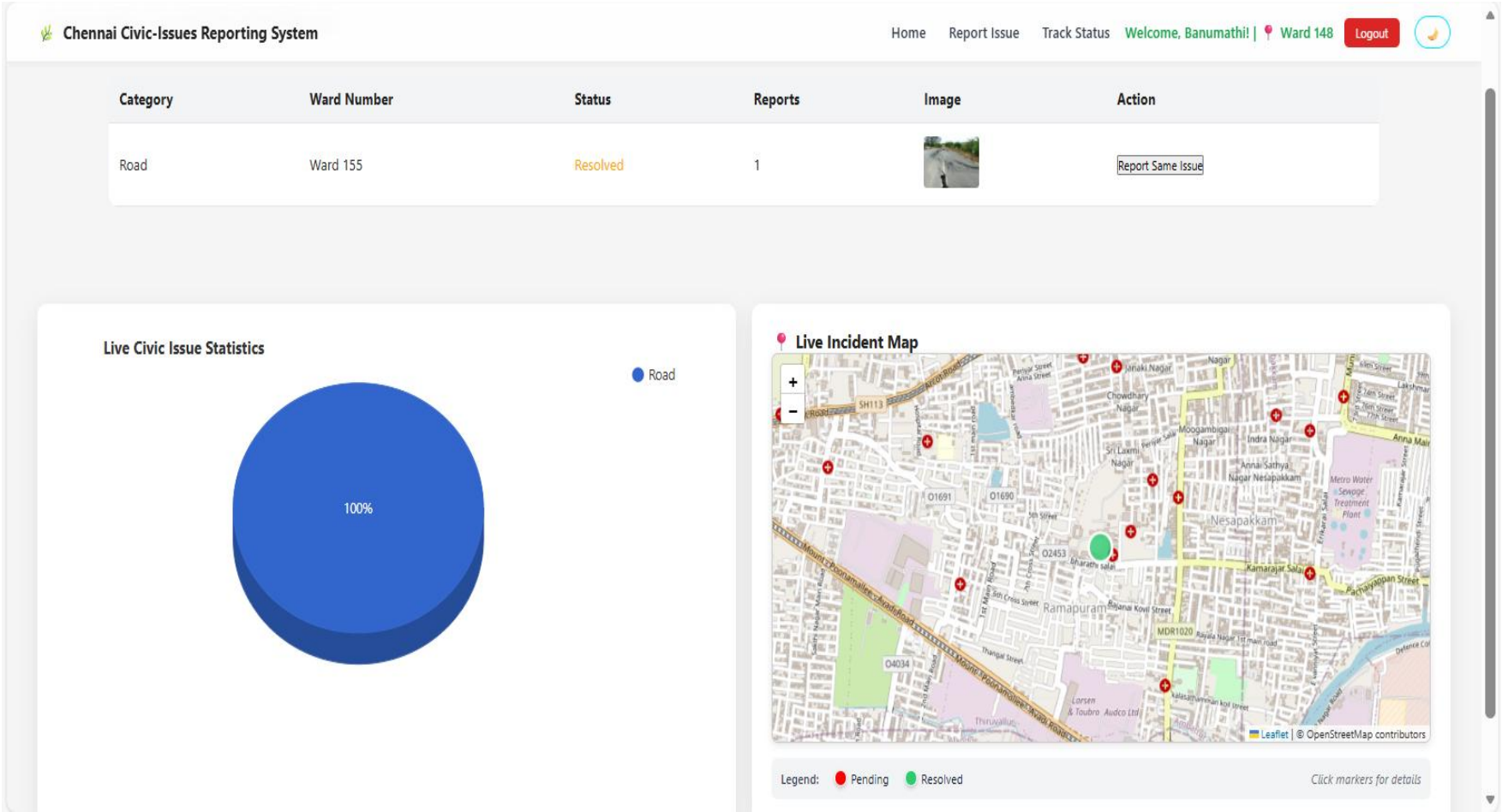
APPLICATION SCREENSHOT

Admin Page



APPLICATION SCREENSHOT

Admin Page



ADVANTAGES

- Easy reporting of civic issues.
- Improves communication between citizens and authorities.
- Provides transparent complaint tracking.
- Maintains digital records of reported issues.

LIMITATIONS

- Requires internet connectivity.
- Developed as a Website not as a Mobile App
- System performance depends on admin responsiveness.
- Large number of complaints may require system scaling.

FUTURE ENHANCEMENTS

- Mobile application integration.
- Notification and Alert System should be Enabled.
- AI-based issue prioritization.
- Integration with government smart city platforms.

CONCLUSION

- The system provides a digital platform for reporting civic issues.
- It improves transparency and communication.
- Authorities can efficiently monitor and resolve problems.
- This contributes to better city management.

THANK YOU